

Let Time Tell: Identification and Gaussian Process Estimation for Interrupted Time Series

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September 25, 2026

Abstract

We study causal inference in interrupted time series designs where a treatment affects every unit simultaneously, so that the contemporaneous controls used by difference-in-differences and synthetic control are unavailable and the counterfactual must be extrapolated from a unit's own pre-treatment history. We establish identification within the potential outcomes framework and estimate the counterfactual using Gaussian process (GP) regression. Rather than committing to a single best-fitting trend, the GP allows for alternative trajectories compatible with the pre-treatment history and widens its intervals where extrapolation magnifies their divergence. Connecting the GP to reproducing kernel Hilbert space theory, we derive a bias decomposition that isolates the learning error associated with extrapolation and establish a sharp worst-case bound on this error, showing how the GP posterior variance provides extrapolation-aware uncertainty quantification. The method is illustrated with calibrated simulations and an analysis of handgun purchases after the Supreme Court's *Heller* decision, a universal treatment whose practical effect concentrates in a single jurisdiction. An R package, *gpss*, implements the approach.

Keywords: causal inference, extrapolation, Gaussian process regression, interrupted time series, machine learning, reproducing kernel Hilbert space

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1 Introduction

The U.S. Supreme Court’s decision in *District of Columbia v. Heller* (2008) recognized an individual right to keep and bear firearms under the Second Amendment and overturned a 1975 D.C. law that had effectively prohibited private handgun ownership (Winkler, 2011). Did the Supreme Court ruling affect handgun purchases in D.C. and if so, by how much? Answering this question proves difficult because, although the immediate regulatory change only occurred in D.C., the ruling was a nationwide legal event that could also affect handgun purchases in other states. We therefore cannot simply regard other states as untreated controls. Thus, approaches that seek to estimate counterfactuals based on what happens in other, putatively untreated states (e.g. Difference-in-Differences (DiD) design (Callaway and Sant’Anna, 2021; Card and Krueger, 1994) and the synthetic control (SC) method (Abadie et al., 2010; Abadie and Gardeazabal, 2003)) are not applicable. Investigators may also seek to leverage an immediate before-after comparison within D.C. to identify the effect within D.C. as in the regression discontinuity in time (RDiT) design (e.g. Hausman and Rapson, 2018). However, such an approach would be an aid to identifying the effect only at the moment the event occurs, whereas we expect the decision’s impact is not entirely instantaneous, and estimating the effect over time is also of interest. We thus need a method for constructing a suitable counterfactual series absent the ruling over a wider post-ruling period, and the only available source is D.C.’s own pre-ruling history.

Many events of interest to social scientists have this structure. A war, a federal policy, or a pandemic reaches every unit at once. In this challenging but not uncommon circumstance, a natural strategy is the interrupted time series (ITS) design (Box and Tiao, 1975; Campbell and Stanley, 1963): learn the pre-event history of the target unit, project forward what would have happened, and read the effect off the gap between the projection and what actually happened. Such an extrapolation-heavy strategy faces several major threats to its validity. First, it assumes that no other event concurrent with the treatment moved the outcome. A rise in handgun purchases in D.C. cannot be attributed to the *Heller* ruling if another deregulatory policy or law took effect in the study period. Second, the projection depends on the model chosen for the pre-event history, and different models that fit the past equally well may predict very different futures. The pre-

event history alone, without a model, cannot rule out subsequent drift driven by factors unseen in that history. Third, the uncertainty of the effect estimate grows the further the projection extends beyond the event, because estimation errors are likely correlated over time and accumulate. Whatever would have moved handgun purchases in D.C. absent the ruling carries over from one month to the next, and thus a projection that is reliable for the month right after the ruling becomes far less credible years later.

A standard method for the ITS design is segmented regression, which fits a linear or polynomial trend to the whole series and adds a dummy variable for the post-event period, whose coefficient gives the effect estimate (Bernal et al., 2017). One drawback most relevant to this paper is that it commits to a single functional form for the trajectory and conditions inference on that choice. Many functions may explain the pre-event data equally well and yet diverge after the event, and the divergence amplifies with temporal distance, which we call extrapolation uncertainty. Segmented regression, and other flexible variants that commit to a single best-fitting function, fail to account for this source of uncertainty inherent in the design. Their confidence intervals may technically widen with the forecast horizon, but only conditional on the chosen functional form; they do not capture uncertainty over alternative trajectories that fit the pre-event history equally well. Adding autoregressive errors lets the interval widen somewhat with the horizon, but that widening comes from the noise around the chosen function, not from the possibility that the function itself drifts a lot in the future.

We therefore need a principled way of quantifying uncertainty that *prices* the extrapolation, so that moving farther beyond the observed data comes with greater uncertainty when plausible trajectories begin to disagree. This paper proposes Gaussian process (GP) regression for the ITS design, which provides such extrapolation-aware inference naturally. The reason is simple: rather than committing to one function, the GP considers a range of trajectories consistent with the pre-event data. The kernel determines how smooth or variable those trajectories may be, and the projection and its uncertainty reflect their center and spread. Near the event, the trajectories tend to agree; farther away, they can diverge drastically, widening the interval. This flexibility is not free, however. The GP cannot tell us which kinds of extrapolation are plausible; that judgment is encoded in the choice of kernel, which determines the space of trajectories the model is willing to

consider.

We next ask whether this widening uncertainty band has a formal connection to the extrapolation problem itself. To do so, we characterize the sources of bias that arise in the ITS design, including the gap between the best counterfactual allowed by the assumed function class and the one the pre-event data can actually recover. We call this gap the learning error. Drawing on reproducing kernel Hilbert space (RKHS) theory, we derive a worst-case bound on this error and show that the bound is extrapolation-aware and tight. In the noiseless case, the GP posterior variance at a post-event point tells us how much trajectories that are indistinguishable over the pre-event series can still disagree there, within the function space induced by the kernel. In simulations calibrated to our application data, the GP intervals keep coverage near the nominal level even with short pre-event histories, while some competing methods lose coverage when the true trajectory drifts away from the fitted line.

Applying the method to monthly handgun background checks in D.C. and 49 other jurisdictions, we find a sharp increase in D.C. immediately after the ruling, well outside the range of counterfactual trajectories supported by the pre-ruling series. In contrast, we find no such effect for other jurisdictions. Two placebo checks provide additional reassurance. Long gun purchases in D.C., which the ruling did not directly concern, show no effect, which suggests that the increase is not simply a general post-ruling pattern. When we move the treatment date to earlier months, the method again finds no effect, suggesting that the result is not the product of an overconfident uncertainty band either.

The rest of the paper proceeds as follows. Section 2 sets up the ITS design as a problem of extrapolation and states the identifying condition. Section 3 introduces the GP estimator and its band. Section 4 develops the bias decomposition and worst-case result underlying the band. Section 5 examines finite-sample properties of the proposed estimator in calibrated simulations. Section 6 applies the method to handgun purchases after *Heller*, and Sections 7–8 situate the estimator among existing approaches and conclude.

2 Identification in Interrupted Time Series Design

Consider a sequence of untreated outcomes before an event of interest, followed by treated outcomes afterward. Without contemporaneous controls, a researcher predicts what would have happened absent treatment by learning from pre-treatment history. Suppose, for example, that the pre-treatment outcomes follow a steady trend with a stable cycle. If that pattern would have continued absent treatment, we can project it forward and treat the gap between the projection and what actually occurs as the treatment effect (Figure 1). The key question is what allows us to carry a pattern learned before treatment into the post-treatment period. Without untreated comparison units, we instead have to rely on the relationship between outcomes and observed characteristics within the unit, and ask whether the relationship learned before treatment can also describe the untreated outcome afterward. This section formalizes this idea.

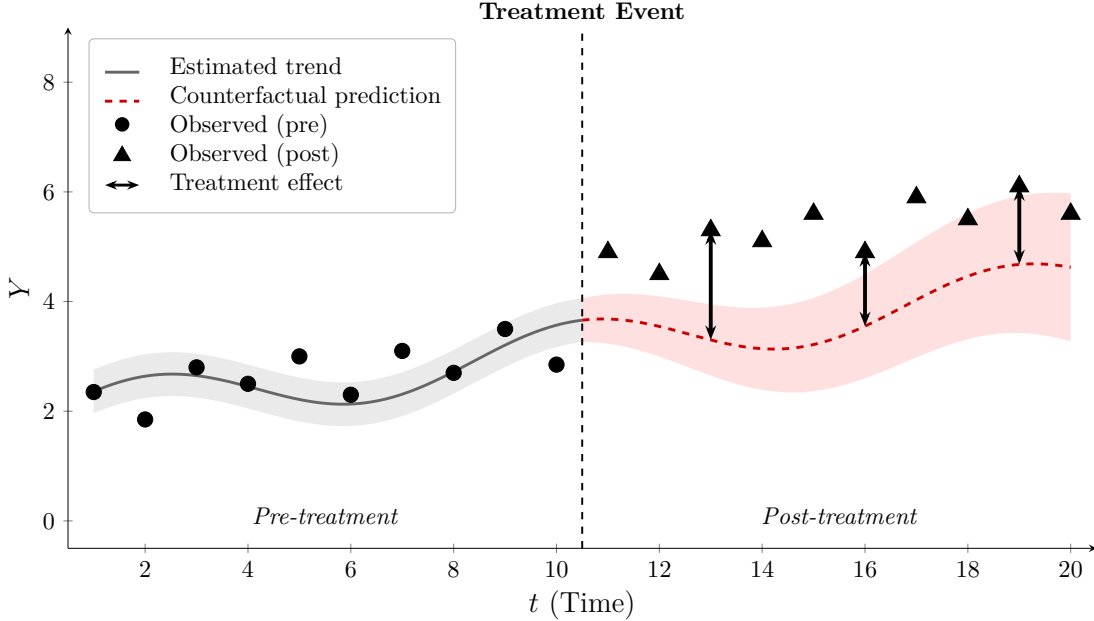


Figure 1. Illustration of Interrupted Time Series Design. *Note:* A flexible trend (solid gray) fitted to pre-treatment outcomes (black dots) is extrapolated as the counterfactual (dashed red), with 95% prediction intervals (shaded) widening with distance from the data. Treatment effects (black arrows) are the gaps between observed outcomes (black triangles) and the counterfactual; the vertical dashed line marks treatment onset.

Setup. Consider a balanced panel of N units observed over T periods. For each unit i and time t , let Y_{it} denote the scalar observed outcome, $D_{it} \in \{0, 1\}$ treatment status, and $\mathbf{X}_{it}$ a vector of observed covariates, including time t itself. Under the potential outcomes framework (Neyman, 1923; Rubin, 1974), let $Y_{it}(1)$ and $Y_{it}(0)$ denote the potential outcomes that would be observed under treatment and control, respectively. Treatment begins simultaneously for all units at time t_0 and persists thereafter, so that $D_{it} = \mathbf{1}(t \geq t_0)$.

Two standard assumptions allow us to use the pre-treatment series as untreated outcomes. *Consistency* links observed and potential outcomes:

$$Y_{it} = D_{it} Y_{it}(1) + (1 - D_{it}) Y_{it}(0). \quad (1)$$

We also require that treatment not affect outcomes before its onset (*no anticipation*):

$$Y_{it}(1) = Y_{it}(0) \quad \text{and} \quad \mathbf{X}_{it}(1) = \mathbf{X}_{it}(0), \quad t < t_0. \quad (2)$$

Together, consistency and no anticipation imply $Y_{it} = Y_{it}(0)$ for $t < t_0$, allowing us to learn the untreated relationship from the observed pre-treatment history. If behavior had already changed in response to the approaching treatment, that history would no longer provide a valid untreated baseline. Expectations that were already present and stable, however, may be part of the relationship learned from the pre-treatment data. The placebo checks in Section 3.3 look for evidence of such pre-treatment instability, and Supplementary Material D.3 discusses the concern in each application. For the covariates, we likewise require $\mathbf{X}_{it^*}(1) = \mathbf{X}_{it^*}(0)$ after treatment so that the observed covariates can be used to construct the untreated counterfactual. This holds by construction when $\mathbf{X}_{it}$ consists of time and calendar indicators such as month or quarter.

Because treatment begins simultaneously for all units, there are no untreated units to be contaminated (i.e., spillovers) from treated units. Interactions among treated units may still affect outcomes, however. The causal quantities below therefore capture total effects, including any such interactions (Cerqua et al., 2023).

Before t_0 , we observe untreated outcomes and use them to learn the counterfactual relationship. After t_0 , we observe treated outcomes and must predict what their untreated counterparts would have been. Let $Y_{i,\text{pre}} = (Y_{i1}, \dots, Y_{i,t_0-1})^\top$ and $\mathbf{X}_{i,\text{pre}} = (\mathbf{X}_{i1}, \dots, \mathbf{X}_{i,t_0-1})^\top$ denote the pre-treatment

outcomes and covariates. At a post-treatment period $t^* \geq t_0$, we observe Y_{it^*} and $\mathbf{X}_{it^*}$.

Structural Model. We next allow the systematic component of the untreated outcome to depend on both observed and unobserved factors. Let $\mathbf{U}_{it}$ collect the unobserved time-varying attributes that may affect outcomes. We write

$$Y_{it}(0) = h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) + \varepsilon_{it}, \quad E[\varepsilon_{it} \mid \mathbf{X}_{it}, \mathbf{U}_{it}] = 0, \quad (3)$$

where the structural function $h_i : \mathcal{X} \times \mathcal{U} \rightarrow \mathbb{R}$ governs how observables and unobservables jointly determine the systematic component of the untreated outcomes, and ε_{it} captures idiosyncratic variation around it. Note that h_i is unit-specific, allowing the same observed and unobserved conditions to affect different units differently. The distinction between $\mathbf{U}_{it}$ and ε_{it} will matter for identification. Unobserved factors in $\mathbf{U}_{it}$ may systematically shift the untreated outcome, whereas ε_{it} has mean zero conditional on both observed and unobserved factors. Mean-zero measurement error may therefore be included in ε_{it} , while systematic changes in measurement or reporting belong in the systematic component (see Supplementary Material D.3).

Target Function. Because $\mathbf{U}_{it}$ is unobserved, h_i itself is not available for prediction. Instead, we target the mean untreated outcome conditional on the observed covariates. We denote this function by $g_i : \mathcal{X} \rightarrow \mathbb{R}$ as

$$g_i(\mathbf{X}_{it}) := E[Y_{it}(0) \mid \mathbf{X}_{it}] = \int h_i(\mathbf{X}_{it}, \mathbf{u}) dP(\mathbf{U}_{it} = \mathbf{u} \mid \mathbf{X}_{it}). \quad (4)$$

Although we cannot observe $\mathbf{U}_{it}$, we can define the mean untreated outcome by averaging over its possible values. Importantly, this average is taken conditional on $\mathbf{X}_{it}$. Unobserved factors may be correlated with the observed covariates, so their distribution can differ across covariate values. The resulting function g_i is therefore the conditional expectation that a well-specified regression of $Y_{it}(0)$ on $\mathbf{X}_{it}$ would recover. In practice, the problem is to learn g_i from the pre-treatment history and then evaluate it at the post-treatment input $\mathbf{X}_{it^*}$. We denote the target by $g_i(\mathbf{X}_{it^*})$ and its estimate by $\hat{g}_i(\mathbf{X}_{it^*})$.

Key Identification Assumption. Defining g_i gives us a target, but it does not by itself justify carrying the relationship learned before treatment into the post-treatment period. The concern is that unobserved conditions may affect the untreated outcome in ways not captured by $\mathbf{X}_{it}$. A concurrent event, for example, could shift the untreated outcome after treatment even if the pre-treatment relationship fit well. We therefore require that, once the observed covariates are known, unobserved factors provide no additional information about the mean untreated outcome.

Assumption 1 (Mean Sufficiency). *For all units i and all periods t :*

$$\mathbb{E}[Y_{it}(0) \mid \mathbf{X}_{it}, \mathbf{U}_{it}] = \mathbb{E}[Y_{it}(0) \mid \mathbf{X}_{it}].$$

Mean Sufficiency does not require the unobserved factors to be constant or unrelated to the observed covariates. It requires only that, once $\mathbf{X}_{it}$ is given, knowing $\mathbf{U}_{it}$ does not further change the mean untreated outcome. Individual outcomes may still fluctuate around that mean through ε_{it} . We can see the restriction more directly by adding and subtracting g_i :

$$Y_{it}(0) = \underbrace{g_i(\mathbf{X}_{it})}_{\text{observable mean}} + \underbrace{[h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) - g_i(\mathbf{X}_{it})]}_{\text{unobservable deviation}} + \underbrace{\varepsilon_{it}}_{\text{noise}}. \quad (5)$$

The middle term captures systematic variation in the untreated outcome that remains unexplained by the observed covariates. Although this term averages to zero conditional on $\mathbf{X}_{it}$, it need not be zero for a particular unit-period. Mean Sufficiency rules out this remaining mean shift by requiring $h_i(\mathbf{X}_{it}, \mathbf{U}_{it}) = g_i(\mathbf{X}_{it})$ almost surely. If the restriction fails, $h_i - g_i$ becomes the identification error isolated in Section 4.

Mean Sufficiency restricts the conditional mean rather than the entire distribution of untreated outcomes. Unobserved factors may still affect other features of the distribution, such as its variance, as long as they do not change the conditional mean. In this sense, the assumption is weaker than the conditional-independence form of selection-on-observables (Hazlett and Xu, 2018). The analogy to selection-on-observables also clarifies what replaces the missing controls. In a cross-sectional setting, observed covariates allow researchers to make comparisons across units. Here, we instead learn the relationship between outcomes and covariates from the same unit’s pre-treatment history and project that relationship forward to construct the counterfactual. The critical requirement is

that this relationship remain valid after treatment.

Combining Mean Sufficiency with consistency and no anticipation gives the pre-treatment regression

$$Y_{it} = g_i(\mathbf{X}_{it}) + \varepsilon_{it}, \quad E[\varepsilon_{it} \mid \mathbf{X}_{it}] = 0. \quad t < t_0, \quad (6)$$

Thus, g_i is the conditional expectation function (CEF) of the observed pre-treatment outcome given $\mathbf{X}_{it}$. This is the relationship we will learn from the pre-treatment series and extrapolate to post-treatment inputs.

Estimands. We define the causal estimands beginning at the unit-period level. At a post-treatment period, we observe the treated outcome but not the same unit's realized untreated outcome. The unit- and period-specific treatment effect is $\delta_{it^*} = Y_{it^*}(1) - Y_{it^*}(0)$. Because $Y_{it^*}(0)$ is not observed, δ_{it^*} is not identified without further strong assumptions. We therefore target its conditional-mean counterpart,

$$\tau_{it^*} = Y_{it^*}(1) - g_i(\mathbf{X}_{it^*}), \quad (7)$$

where $g_i(\mathbf{X}_{it^*}) = \mathbb{E}[Y_{it^*}(0) \mid \mathbf{X}_{it^*}]$. This replaces the unrecoverable realization with its conditional mean given observables, the quantity that standard ITS methods actually estimate. Aggregate estimands average or accumulate the unit-period effects,

$$\bar{\tau}_{t^*} = \frac{1}{N} \sum_{i=1}^N \tau_{it^*}, \quad \bar{\tau}_i = \frac{1}{T - t_0 + 1} \sum_{s=0}^{T-t_0} \tau_{i,t_0+s}, \quad \tau_{i,\text{cum}} = \sum_{t^*=t_0}^T \tau_{it^*}, \quad (8)$$

the average effect across units at t^* , the average post-treatment effect for unit i , and its cumulative effect, respectively. Identification therefore turns on whether g_i , learned from the pre-treatment history, can be recovered at the post-treatment inputs.

Theorem 1 (Identification of Treatment Effects). *Under consistency (1), no anticipation (2), and Assumption 1, the treatment effect at $t^* \geq t_0$ is identified as*

$$\tau_{it^*} = Y_{it^*} - g_i(\mathbf{X}_{it^*}),$$

where g_i coincides with the CEF of the observed pre-treatment data, $g_i(\mathbf{x}) = \mathbb{E}[Y_{it} \mid \mathbf{X}_{it} = \mathbf{x}]$ for

$t < t_0$.

A proof is given in Supplementary Material A. The result leaves one central task: recovering the untreated counterfactual mean $g_i(\mathbf{X}_{it^*})$. The pre-treatment data reveal the relationship over the observed inputs, but because time itself is included in $\mathbf{X}_{it}$, evaluating g_i after treatment requires extending that relationship beyond the observed period. This is where extrapolation uncertainty enters. Different trajectories may agree closely, or even exactly, throughout the pre-treatment period and still diverge afterward. We refer to the range of such disagreement permitted by the assumed function class as *extrapolation uncertainty*. Section 4 formalizes this idea after introducing the GP and its associated function space.

The main threat to Mean Sufficiency assumption is inherent to the ITS design rather than to any particular estimator. An unobserved factor may be stable throughout the pre-treatment period but begin to affect the untreated outcome differently after treatment. A model could then fit the observed history well while mistaking this new influence for a treatment effect. In our *Heller* application, for example, the possibility of election-related anxiety about gun restrictions motivates ending the analysis before the November election (Section 6). Such threats generally become harder to rule out over longer horizons, making shorter post-treatment windows more credible when concurrent changes are a concern. Placebo checks can reveal instability already present before treatment, but they cannot rule out a break that begins only afterward.

3 Proposed Methodology: Gaussian Process Estimation for ITS

The identification result tells us what must be recovered: the counterfactual mean $g_i(\mathbf{X}_{it^*})$ at a post-treatment input. The remaining problem is one of estimation—how to learn this function from the pre-treatment history and extend it beyond the observed data. We first describe the general estimation procedure and then introduce Gaussian process regression as our proposed approach (see [Rasmussen et al. \(2006\)](#) for a textbook treatment).

3.1 A General Estimation Procedure

Theorem 1 reduces estimation to a straightforward function learning problem. We learn the untreated relationship $g_i(\cdot)$ from the pre-treatment series, extend it to each post-treatment input, and compare the resulting counterfactual with the observed outcome.

More formally, for each unit i , we (a) estimate g_i from the pre-treatment data $(\mathbf{X}_{i,\text{pre}}, Y_{i,\text{pre}})$ to obtain $\hat{g}_i$; (b) predict the counterfactual outcome $\hat{Y}_{it^*}(0) = \hat{g}_i(\mathbf{X}_{it^*})$, for each post-treatment period $t^* \geq t_0$; (c) estimate the treatment effect as $\hat{\tau}_{it^*} = Y_{it^*} - \hat{g}_i(\mathbf{X}_{it^*})$; and (d) aggregate these effects to the summary estimands as needed. Unlike standard ITS specifications that impose a step, pulse, or gradual treatment response (McDowall et al., 2019; Schaffer et al., 2021), this procedure places no *a priori* restriction on the shape of the treatment effect over time. The pattern of τ_{it^*} is instead allowed to emerge period by period. At this stage, nothing requires g_i to be estimated by a GP; any estimator can be used without changing the causal interpretation above.

The harder question is how reliably the pre-treatment data determine g_i at a post-treatment input. Prediction now requires extrapolation, so even an estimator that fits the observed history well may recover the counterfactual poorly beyond it. As a useful benchmark, we would like the estimated counterfactual to equal its target on average:

$$\mathbb{E}[\hat{g}_i(\mathbf{X}_{it^*}) \mid \mathbf{X}_{i,\text{pre}}, \mathbf{X}_{it^*}] = g_i(\mathbf{X}_{it^*}). \quad (9)$$

We consider this condition as a benchmark rather than an assumption. With finite and noisy pre-treatment data, estimation and regularization can leave a gap between $\hat{g}_i$ and g_i at the extrapolation point. We refer to this as learning error and return to it in Section 4.

3.2 Gaussian Process Estimation

GP regression models the relationship among function values through a covariance kernel. The kernel encodes which inputs are expected to produce similar outcomes and how quickly that similarity decays as the inputs become more different. In our setting, this provides a flexible way to learn the pre-treatment trajectory while making the assumptions used to extend it beyond the observed history explicit. Just as importantly, the GP gives a distribution over possible counterfactual con-

tinuations rather than only a single prediction. We model the untreated potential outcome for unit i at time t as

$$Y_{it}(0) = f_i(\mathbf{X}_{it}) + \varepsilon_{it}, \quad f_i \sim \mathcal{GP}(0, k), \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2). \quad (10)$$

Here f_i is a unit-specific latent function and k is the covariance kernel that governs its structure. The kernel matrix $\mathbf{K}_{\text{pre}}$, with entries $K_{tt'} = k(\mathbf{X}_{it}, \mathbf{X}_{it'})$, collects these relationships across the pre-treatment inputs. We use the GP posterior mean to estimate g_i . Gaussian noise is assumed for posterior computation and hyperparameter estimation, but the identification result and subsequent bias decomposition do not require normality.

Posterior Prediction and Inference. We fit the GP using only the pre-treatment observations and then condition on that history to predict the untreated trajectory after treatment. The posterior mean gives the counterfactual prediction, while the posterior predictive covariance describes how uncertain that continuation remains. Under the GP model, the pre- and post-treatment outcomes are jointly Gaussian, yielding the predictive distribution (Rasmussen et al., 2006):

$$\mathbf{Y}_{i,\text{post}}(0) \mid Y_{i,\text{pre}}, \mathbf{X}_{i,\text{pre}}, \mathbf{X}_{i,\text{post}} \sim \mathcal{N}(\boldsymbol{\mu}_{\text{post}}, \boldsymbol{\Sigma}_{\text{post}}),$$

$$\boldsymbol{\mu}_{\text{post}} = \mathbf{K}_{\text{post,pre}}(\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} Y_{i,\text{pre}}, \quad (11)$$

$$\boldsymbol{\Sigma}_{\text{post}} = \mathbf{K}_{\text{post}} + \sigma^2 I - \mathbf{K}_{\text{post,pre}}(\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} \mathbf{K}_{\text{pre,post}}, \quad (12)$$

where $\mathbf{K}_{\text{post,pre}}$ has (t^*, t) -entry $k(\mathbf{X}_{it^*}, \mathbf{X}_{it})$ and $\mathbf{K}_{\text{post}}$ has (t^*, t^{**}) -entry $k(\mathbf{X}_{it^*}, \mathbf{X}_{it^{**}})$. Writing $\mathbf{k}_{t^*}^\top$ for the covariance between the post-treatment input and the pre-treatment inputs, the posterior mean at t^* is $\hat{g}_i(\mathbf{X}_{it^*}) = \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} Y_{i,\text{pre}}$. This is the counterfactual used in the treatment-effect estimate. The corresponding diagonal element of $\boldsymbol{\Sigma}_{\text{post}}$ gives its predictive variance. Section 4 returns to this variance and shows how it relates to learning error.

Kernel Specification. The kernel determines which inputs the model regards as similar, and therefore which observations are informative about one another. In a simple time-series setting, nearby periods may be expected to have more similar function values than periods far apart in time. The kernel therefore makes explicit what we are willing to assume about over-time comparability:

different kernel choices imply different views about how quickly the untreated trajectory may change as observations become farther apart.

One useful building block is the Gaussian kernel,

$$k_{\text{Gaussian}}(\mathbf{X}, \mathbf{X}') = \exp\left(-\frac{\|\mathbf{X} - \mathbf{X}'\|^2}{b}\right), \quad (13)$$

which can approximate a broad class of continuous functions (Micchelli et al., 2006). In our setting, it encodes the assumption that temporally proximate observations with similar covariate values have similar function values. The same kernel also determines the function space used in the theoretical results of Section 4; we defer that RKHS interpretation until then.

Combining Kernels. A Gaussian kernel alone is not well suited to every feature of a time series. Because it is stationary, observations far from the pre-treatment data receive little covariance, predictions tend back toward the prior mean, and predictive uncertainty approaches a prior variance as the ceiling. This behavior is poorly suited to a series with a persistent long-run trend. For the time series considered here, we want the kernel to accommodate three common features separately: local smooth variation, recurring seasonality, and longer-run trend. Sums of valid covariance functions are valid covariance functions (Rasmussen et al., 2006), so components can be assembled to suit an application (Duvenaud et al., 2013); our working form is

$$k_{\text{combined}}(\mathbf{X}, \mathbf{X}') = k_{\text{Gaussian}}(\mathbf{X}, \mathbf{X}') + k_{\text{periodic}}(\mathbf{X}, \mathbf{X}') + k_{\text{linear}}(\mathbf{X}, \mathbf{X}'). \quad (14)$$

The periodic kernel $k_{\text{periodic}}(\mathbf{X}, \mathbf{X}') = \exp(-2 \sin^2(\pi|\mathbf{X} - \mathbf{X}'|/p) / (b/2))$ captures seasonal patterns with period p , assigning maximum covariance to time points separated by integer multiples of p (MacKay et al., 1998). The linear kernel $k_{\text{linear}}(\mathbf{X}, \mathbf{X}') = \mathbf{X}^\top \mathbf{X}'$ models trends that extend beyond the observed data, so predictive uncertainty grows with distance from the pre-treatment support. We standardize the outcome and give the three components equal amplitude in our working specification (Cho et al., 2026). Because this composition is itself an extrapolation assumption, researchers should examine sensitivity to alternative kernel specifications when other continuations are substantively plausible.

Hyperparameter Selection. We standardize continuous covariates and outcomes within each unit and one-hot encode categorical covariates, so that the kernel operates on comparable scales; Supplementary Material F gives implementation details. The length-scale b controls how rapidly covariance decays with distance: larger values favor smoother functions, while smaller values allow more rapid local variation. We select b using the variance-maximization rule proposed in [Hartman et al. \(2025\)](#), $\hat{b} = \operatorname{argmax}_b \mathbb{V}(K_{ij})$ for $i \neq j$, where K_{ij} are the off-diagonal entries of $\mathbf{K}_{\text{pre}}$. Intuitively, this avoids two uninformative extremes: a bandwidth so small that nearly every pair of observations looks unrelated, and one so large that nearly every pair looks the same. Because the rule depends only on the covariates, it also keeps the choice of length-scale separate from the outcome data, keeping the design and analysis stages separate ([Chattopadhyay and Zubizarreta, 2023](#)). We assess sensitivity to b in Supplementary Material F. The period p is set from the natural cycle of the data, such as $p = 12$ for monthly observations, or from the detrended periodogram when the cycle is unknown.

Noise Variance and Regularization. With the length-scale fixed at $\hat{b}$, σ^2 controls how much of the pre-treatment variation is treated as noise rather than meaningful movement in the underlying function. Larger values produce a smoother fitted trajectory, while smaller values follow the observed series more closely. We estimate σ^2 by marginal likelihood: $\hat{\sigma}^2 = \operatorname{argmax}_{\sigma^2} \log p(Y_{i,\text{pre}} | \mathbf{X}_{i,\text{pre}}, \hat{b}, \sigma^2)$.

The same parameter also regularizes the fitted function. This is clear from the equivalent kernel ridge regression problem,

$$\hat{g}_i = \operatorname{argmin}_{f \in \mathcal{H}_k} \sum_{t < t_0} (Y_{it} - f(\mathbf{X}_{it}))^2 + \sigma^2 \|f\|_{\mathcal{H}_k}^2. \quad (15)$$

Larger σ^2 therefore places a stronger penalty on functions that are complex under the chosen kernel, trading closer fit to the pre-treatment observations for greater smoothness. In our procedure, b is chosen from the covariates first, and σ^2 is then estimated from the pre-treatment outcomes with b fixed.

3.3 Placebo Checks

Our identification strategy uses the pre-treatment relationship to predict untreated outcomes after treatment. We can assess how well this forecasting strategy works by treating earlier pre-treatment periods as held-out targets, where the true treatment effect is zero (Eggers et al., 2023). This provides a within-unit temporal analogue of a pre-trends check in DiD design.

The placebo check works by pretending the treatment started earlier than the actual treatment onset (cf. Liu et al., 2024). Take the last M pre-treatment periods $\{t_0 - M, \dots, t_0 - 1\}$ as a placebo post-treatment window, for instance matching the length of the actual one, and define relative time $m = -M, \dots, -1$, with $m = -1$ the period immediately before onset. For each m we fit the GP on all periods before $t_0 + m$ and form the placebo estimate

$$\hat{\tau}_{i,m}^{\text{placebo}} = Y_{i,t_0+m} - \hat{g}_i^{(t_0+m)}(\mathbf{X}_{i,t_0+m}), \quad m = -M, \dots, -1, \quad (16)$$

where $\hat{g}_i^{(t_0+m)}$ is the GP posterior mean trained on $\{1, \dots, t_0 + m - 1\}$. Since no treatment occurs at these placebo dates, the estimated effects should remain near zero and their intervals should cover zero at the stated rate. These are one-step-ahead predictions, matching the forecasting problem at the first actual post-treatment period. Good placebo performance is therefore reassuring about short-horizon prediction, but it does not guarantee accuracy farther into the post-treatment period.

The placebo is primarily a falsification check on pre-treatment forecasting performance. Systematic placebo errors may indicate either a poorly specified kernel or instability in the relationship being extrapolated. A clean placebo, however, cannot establish Mean Sufficiency: an unobserved disturbance that appears only after treatment would remain undetected. Temporal placebos can also be supplemented with placebo outcomes where available (see Felton, 2023). Supplementary Material E reports these checks for the empirical application.

3.4 Aggregation for Panel Data

With N units observed over common periods, we fit a separate GP to each unit’s pre-treatment series and obtain unit-specific counterfactuals $\hat{g}_i(\mathbf{X}_{it^*})$. This allows temporal trajectories to differ freely across units, although it does not borrow strength across them. The unit-specific treatment

effect is $\hat{\tau}_{it^*} = Y_{it^*} - \hat{g}_i(\mathbf{X}_{it^*})$. We then aggregate these estimates across units or post-treatment periods:

$$\hat{\tau}_{t^*} = \frac{1}{N} \sum_{i=1}^N \hat{\tau}_{it^*}, \quad \hat{\tau}_i = \frac{1}{T - t_0 + 1} \sum_{s=0}^{T-t_0} \hat{\tau}_{i,t_0+s}, \quad \hat{\tau}_{i,\text{cum}} = \sum_{t^*=t_0}^T \hat{\tau}_{it^*}. \quad (17)$$

These correspond to the average effect across units at t^* , the average post-treatment effect for unit i , and its cumulative effect. Because the unit-specific GPs are fitted independently, the model also treats their posterior predictive distributions as independent conditional on the data. This makes aggregation straightforward, but it could understate uncertainty when prediction errors are correlated across units because of common shocks or spatial dependence (cf. [Ben-Michael et al., 2023](#)). In particular, positive cross-unit dependence would make aggregate intervals too narrow.

4 Theoretical Properties of the Gaussian Process Estimator

The GP gives us a counterfactual prediction, but several distinct sources of error can make that prediction differ from the untreated outcome we want to recover. These errors arise from distinct sources: identification, restrictions imposed by the kernel’s function class, and estimation from finite and noisy pre-treatment data. We first decompose these sources and then focus on the one most directly tied to extrapolation: learning error. The main result shows how the GP posterior variance controls this error.

4.1 Setup and the Four Functions

We suppress the unit subscript i throughout. Four functions mark the steps between the untreated outcome and what we ultimately estimate. The structural function $h(\mathbf{X}_t, \mathbf{U}_t)$ describes the systematic part of the untreated outcome when both observed and unobserved factors are known. Averaging over the unobserved factors gives $g(\mathbf{X}_t) = \mathbb{E}[Y_t(0) \mid \mathbf{X}_t]$, the counterfactual mean we would like to recover. The kernel restricts the class of functions available to the estimator; g^* denotes the best approximation to g within that class. Finally, $\hat{g}$ is the function we actually estimate from the finite pre-treatment data.

We now turn to the distinction between g and g^* . Even if g is identified, the estimator can only recover functions that are sufficiently compatible with the chosen kernel. In this sense, the kernel does more than determine how observations are related; it also restricts the class of counterfactual trajectories available to the estimator. Formally, each kernel induces a reproducing kernel Hilbert space, $\mathcal{H}_k$, which we can think of as the collection of functions the kernel can represent. The RKHS norm $\|f\|_{\mathcal{H}_k}$ measures how costly a function is under that representation: functions that follow patterns naturally accommodated by the kernel have smaller norms, while less compatible functions have larger ones. We restrict attention to functions within a complexity budget B , $\mathcal{F}_B = \{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq B\}$, so that larger values of B permit a broader set of functions. Within this class, g^* is the best approximation to the true conditional mean g over the covariate support relevant to the analysis. Let $P_{\mathbf{X}}$ denote a reference distribution over this support. Then

$$g^* = \operatorname{argmin}_{f \in \mathcal{F}_B} \|f - g\|_{L^2(P_{\mathbf{X}})}. \quad (18)$$

If $g \in \mathcal{F}_B$, then $g^* = g$ and there is no approximation error. Otherwise, some discrepancy remains regardless of how much pre-treatment data we observe: additional data may help us learn g^* , but they cannot recover features of g that the assumed function class does not allow. We return to the technical role of the norm bound and the approximation properties of the kernel in Supplementary Material B.

The last step is from g^* , the best function available within the assumed class, to $\hat{g}$, the function we can actually learn from the observed pre-treatment data. Even when g^* is a good approximation to g , a finite and noisy history may not contain enough information to recover its value at a post-treatment input. More formally, the GP posterior mean $\hat{g}$ lies, by the representer theorem (Kimeldorf and Wahba, 1970), in the empirical subspace $\mathcal{S}_{\text{pre}} = \text{span}\{k(\cdot, \mathbf{X}_t) : t = 1, \dots, t_0 - 1\} \subset \mathcal{H}_k$. The pre-treatment inputs constrain the plausible function through this subspace. However, the key difficulty is that those inputs may constrain the function well over the observed history while still leaving considerable discretion at a post-treatment point. Finite-sample noise and regularization can further widen the gap between g^* and $\hat{g}$, which we will call learning error. Table 1 summarizes the four functions.

Table 1. Functions in the bias decomposition for counterfactual prediction.

Function	Definition	Governed by
$h(\mathbf{X}_t, \mathbf{U}_t)$	Structural function	Observables + unobservables
$g(\mathbf{X}_t)$	CEF: $\mathbb{E}[Y_t(0) \mid \mathbf{X}_t]$	Observables only
$g^*(\mathbf{X}_t)$	Best $\mathcal{F}_B$ approximation of g	Kernel + complexity budget
$\hat{g}(\mathbf{X}_t)$	GP estimate from pre-treatment data	Kernel + observed data

4.2 Bias Decomposition

With these four functions in place, we can analyze the different ways counterfactual prediction can fail. The problem may arise because unobserved factors shift the untreated outcome in ways not captured by $\mathbf{X}_t$; because the function class implied by the kernel does not represent the true conditional mean well; because the finite pre-treatment series does not reveal enough about the function at the post-treatment input; or simply because the realized untreated outcome contains idiosyncratic noise. These correspond, respectively, to identification, approximation, learning, and noise error.

To display all four sources at once, it is useful to compare the estimated effect with the realized treatment effect δ_{t^*} , rather than with our primary conditional-mean estimand τ_{t^*} first:

$$\hat{\tau}_{t^*} - \delta_{t^*} = Y_{t^*}(0) - \hat{g}(\mathbf{X}_{t^*}) \quad (19)$$

$$= \underbrace{[h(\mathbf{X}_{t^*}, \mathbf{U}_{t^*}) - g(\mathbf{X}_{t^*})]}_{\text{I: identification}} + \underbrace{[g(\mathbf{X}_{t^*}) - g^*(\mathbf{X}_{t^*})]}_{\text{II: approximation}} + \underbrace{[g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})]}_{\text{III: learning}} + \underbrace{\varepsilon_{t^*}}_{\text{IV: noise}}. \quad (20)$$

Although the decomposition is written relative to the realized treatment effect δ_{t^*} , our primary estimand τ_{t^*} replaces the realized untreated outcome with its conditional mean. Relative to this estimand, $\hat{\tau}_{t^*} - \tau_{t^*} = g(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})$, so only approximation and learning error remain. Identification error and idiosyncratic noise matter when we ask how the estimate compares with the particular untreated outcome that would have occurred, but not when the target is its conditional mean. This distinction lets us separate limitations of the causal design from limitations of the function class and the data used to learn the counterfactual.

Term I: Identification error. This term captures systematic variation in the untreated outcome that remains attributable to $\mathbf{U}_{t^*}$ after conditioning on $\mathbf{X}_{t^*}$. Mean Sufficiency assumption

rules out this source of error by requiring $h(\mathbf{X}_{t^*}, \mathbf{U}_{t^*}) = g(\mathbf{X}_{t^*})$ almost surely.

Term II: Approximation error. Even when the counterfactual mean g is identified, the function class implied by the kernel may not be rich enough to represent it. The gap $g - g^*$ measures this mismatch. If $g \in \mathcal{F}_B$, the term vanishes; otherwise, it remains regardless of how much pre-treatment data we observe because the limitation lies in the function class rather than in the sample.

Term III: Learning error. A different problem remains even when $g^* = g$: the available pre-treatment data may not determine g^* well at a post-treatment input. The GP estimator must recover the function from a finite and noisy sample, while regularization also favors functions that are less costly under the kernel. The resulting gap between g^* and $\hat{g}$ is the learning error on which our main theoretical result focuses, summarized in Proposition 2.

Term IV: Noise. Finally, the realized untreated outcome contains idiosyncratic variation ε_{t^*} that no counterfactual estimator can predict. Under the structural model (3), this term has conditional mean zero and contributes σ^2 to the outcome variance.

Once the target is set to the conditional-mean effect τ_{t^*} , and Mean Sufficiency holds, conditional bias therefore comes from two distinct questions: whether the chosen function class can represent g , and whether the pre-treatment data can recover the best function available within that class. The following result states this separation explicit.

Proposition 1 (Conditional Bias). *The conditional bias of $\hat{\tau}_{t^*}$ for the primary estimand τ_{t^*} satisfies*

$$\mathbb{E}[\hat{\tau}_{t^*} - \tau_{t^*} \mid \mathbf{X}_{\text{pre}}, \mathbf{X}_{t^*}] = \underbrace{g(\mathbf{X}_{t^*}) - g^*(\mathbf{X}_{t^*})}_{\text{approximation bias}} + \underbrace{\mathbb{E}[g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*}) \mid \mathbf{X}_{\text{pre}}, \mathbf{X}_{t^*}]}_{\text{learning bias}}. \quad (21)$$

Under Assumption 1 and $\mathbb{E}[\varepsilon_{t^} \mid \mathbf{X}_{\text{pre}}, \mathbf{X}_{t^*}] = 0$, the same expression gives the conditional bias relative to the realized effect δ_{t^*} .*

The distinction is useful because the two biases have different sources and therefore different remedies. Approximation bias reflects the choice of function class; observing a longer pre-treatment history does not by itself correct a class that excludes important features of g . Learning bias instead reflects what the available pre-treatment data allow us to recover within that class, including the effects of finite-sample noise and regularization. When $g^* = g$, zero learning bias recovers the

benchmark in (9). We now ask how the GP’s own uncertainty management relates to this remaining learning problem.

4.3 RKHS Interpretation of GP Posterior Variance

We now focus on learning error that we can control. Suppose g belongs to the chosen function class, so that $g^* = g$. Even then, observing the function only before treatment need not determine how it continues afterward: several admissible trajectories may agree throughout the pre-treatment period while implying quite different values at a post-treatment point. The question is how much freedom remains once the pre-treatment observations have been taken into account.

The RKHS representation gives a natural way to answer this question. The pre-treatment inputs constrain some directions of the function space more strongly than others, while a post-treatment input may contain a component that those observations do not represent. The GP posterior variance captures both this unresolved component and the uncertainty introduced by estimating the function from noisy pre-treatment outcomes.

Before turning to the GP estimator, it is useful to define the underlying extrapolation problem in general. Consider two counterfactual trajectories that are both allowed by the chosen function class and agree at every observed pre-treatment input. How far apart could they nevertheless be at t^* ? We call the largest such disagreement *extrapolation uncertainty*, adapting [Shen and Meinshausen \(2025, Definition 1\)](#) to the ITS setting.

Definition 1 (Extrapolation uncertainty). *Let $\mathcal{F}_B = \{f \in \mathcal{H}_k : \|f\|_{\mathcal{H}_k} \leq B\}$ denote the RKHS ball with complexity budget B . For a post-treatment input $\mathbf{X}_{it^*}$, define the extrapolation uncertainty of $\mathcal{F}_B$ as*

$$\omega_B(\mathbf{X}_{t^*}) = \sup\{|f(\mathbf{X}_{t^*}) - f'(\mathbf{X}_{t^*})| : f, f' \in \mathcal{F}_B, f(\mathbf{X}_t) = f'(\mathbf{X}_t) \text{ for all } t < t_0\}. \quad (22)$$

In other words, $\omega_B(\mathbf{X}_{t^*})$ measures how much two admissible counterfactuals can disagree after treatment even though the pre-treatment data cannot distinguish between them. Importantly, this is a property of the function class and the observed input locations, not of a particular estimator.

We can now ask how this unresolved extrapolation appears in the GP prediction (see [Kanagawa](#)

et al. (2025) for a recent treatment of this GP–RKHS connection). As shown in Section 3.2, the GP posterior mean is equivalent to a kernel ridge regression estimator in $\mathcal{H}_k$ (Kimeldorf and Wahba, 1970). It can also be written as a weighted combination of the pre-treatment outcomes, $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_t$, which makes the source of learning error clear. Even if the pre-treatment function values were observed without noise, their weighted combination might fail to reproduce $g^*(\mathbf{X}_{t^*})$; with noisy outcomes, the same weights also carry sampling noise into the prediction. Thus, learning error has two parts: a deterministic reconstruction error and noise propagated from the pre-treatment observations. The next proposition shows that the GP posterior variance accounts for both. For simplicity, we state the result for $B = 1$; the deterministic bound scales linearly with B .

Proposition 2 (GP posterior variance controls the learning error). *Suppose g lies in the unit ball $\mathcal{F}_1$, so that $g^* = g$, and suppose outcomes are observed with noise, $Y_t = g^*(\mathbf{X}_t) + \varepsilon_t$, where ε_t are mean-zero with variance $\sigma^2 > 0$. The GP posterior mean is the weighting estimator $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_t$ with weights $\mathbf{w} = (\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} \mathbf{k}_{t^*}$, and its posterior variance is*

$$\mathbb{V}_{t^*} = k(\mathbf{X}_{t^*}, \mathbf{X}_{t^*}) - \mathbf{k}_{t^*}^\top (\mathbf{K}_{\text{pre}} + \sigma^2 I)^{-1} \mathbf{k}_{t^*}.$$

Let $k_{t^*}^\perp = k(\cdot, \mathbf{X}_{t^*}) - \sum_t w_t k(\cdot, \mathbf{X}_t)$ be the residual of the test feature after its ridge reconstruction from the training-feature span $\mathcal{S}_{\text{pre}} = \text{span}\{k(\cdot, \mathbf{X}_t)\}$, coinciding with the orthogonal-projection residual onto $\mathcal{S}_{\text{pre}}$ as $\sigma^2 \rightarrow 0$. Then the learning error decomposes as

$$g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*}) = \underbrace{\left(g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) \right)}_{\text{deterministic error}} - \underbrace{\sum_t w_t \varepsilon_t}_{\text{noise propagation}},$$

and its two sources are controlled by the two summands of the GP posterior variance:

$$\left| g^*(\mathbf{X}_{t^*}) - \sum_t w_t g^*(\mathbf{X}_t) \right| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k}, \quad \text{Var}\left[\sum_t w_t \varepsilon_t\right] = \sigma^2 \|\mathbf{w}\|^2, \quad \underbrace{\|k_{t^*}^\perp\|_{\mathcal{H}_k}^2 + \sigma^2 \|\mathbf{w}\|^2}_{\text{GP posterior variance}} = \mathbb{V}_{t^*}.$$

Therefore, the first summand of $\mathbb{V}_{t^*}$ gives the squared worst-case bound on the deterministic error and the second gives the variance of the propagated noise.

A proof is given in Supplementary Material B. The proposition gives the posterior variance a simple interpretation. The first component, $\|k_{t^*}^\perp\|_{\mathcal{H}_k}$, measures what the pre-treatment inputs fail to

determine about the post-treatment point within the chosen function space. When this residual is small, trajectories that are compatible with the observed history are also tightly constrained at t^* . When it is large, trajectories can remain difficult to distinguish before treatment yet imply substantially different counterfactual outcomes afterward. The second component, $\sigma^2 \|\mathbf{w}\|^2$, measures the additional uncertainty propagated into the prediction from noise in the pre-treatment outcomes.

This is what makes the bound extrapolation-aware. As the target moves into regions that are less well represented by the pre-treatment inputs, the unrecovered component can grow, and the uncertainty grows with it. Importantly, however, distance here is determined by the kernel’s notion of similarity rather than by calendar time alone. A trajectory that is easy to represent under one kernel may be costly under another, so the same post-treatment point can be well constrained under one set of functional assumptions and weakly constrained under another. Our combined kernel is designed with this in mind, giving trend, seasonality, and local variation separate components in which they can be represented naturally. Supplementary Material B develops this connection further, including the corresponding source-condition refinement. Supplementary Material B develops this relationship formally and gives the corresponding source-condition refinement of this idea.

The connection is particularly clear when the pre-treatment outcomes are observed without noise. In that case, there is no propagated-noise component, so the remaining learning error comes entirely from what the pre-treatment inputs fail to reconstruct the function at t^* . For the unit RKHS ball, Corollary B.1 (Supplementary Material B) gives

$$|g^*(\mathbf{X}_{t^*}) - \hat{g}(\mathbf{X}_{t^*})| \leq \|k_{t^*}^\perp\|_{\mathcal{H}_k} = \sqrt{\mathbb{V}_{t^*}}, \quad (23)$$

where $\mathbb{V}_{t^*}$ denotes the GP posterior variance with $\sigma^2 = 0$. Thus, in the noiseless case, the GP posterior standard deviation gives a worst-case bound on how far the function recovered from the pre-treatment data can lie from the best admissible counterfactual at the post-treatment point.

The bound is also sharp. To see why, suppose two functions in the admissible class fit the pre-treatment history exactly but imply different counterfactual values at t^* . From the pre-treatment data alone, there is no way to know which continuation is the correct one. A single prediction must therefore fall somewhere between these two possibilities, and even the best choice—the midpoint between them—will still be wrong by half the distance between the two values if either function

is the truth. The minimax risk is the smallest worst-case error that any estimator can guarantee over the admissible function class. Supplementary Material B shows that the GP posterior mean attains this limit. More generally, for $\mathcal{F}_B$,

$$\mathfrak{R}_B(\mathbf{X}_{t^*}) = B\|k_{t^*}^\perp\|_{\mathcal{H}_k} = \frac{\omega_B(\mathbf{X}_{t^*})}{2}. \quad (24)$$

The last equality provides the direct connection to extrapolation uncertainty: if two admissible trajectories can differ by as much as $\omega_B(\mathbf{X}_{t^*})$ while remaining indistinguishable before treatment, no estimator can uniformly guarantee an error smaller than half that disagreement. With noisy pre-treatment outcomes, Proposition 2 adds the uncertainty generated by noise propagation. Proofs and further results are given in Supplementary Material B.

These results clarify why GP regression is well suited to the extrapolation problem in ITS. The pre-treatment data may fit several plausible trajectories without telling us whether the untreated outcome would continue linearly or more wildly. The GP cannot determine which continuation is correct, but its kernel makes the assumed class of trajectories explicit, and its posterior variance provides a principled measure of uncertainty within that specification. Allowing more complex continuations, such as quadratic rather than only linear trends, may lead to greater uncertainty and weaker conclusions. The argument can also be read in reverse: if an estimated effect remains distinguishable from zero only under a kernel that restricts the counterfactual to a linear trend, the researcher must defend excluding more flexible alternatives. Comparing results across plausible kernels therefore reveals which extrapolation assumptions are needed to sustain a causal conclusion.

5 Calibrated Simulation Studies

We now examine how the proposed estimator behaves in finite samples, focusing in particular on whether its uncertainty intervals remain well calibrated as the counterfactual is projected beyond the observed history. The simulations are designed to vary both the amount of pre-treatment information available and how closely the untreated trajectory aligns with the function class assumed by the GP.

Setup. We calibrate the data generating process (DGP) to the monthly series in our empirical application (Section 6), which exhibits seasonality and an underlying trend. Untreated outcomes are generated as $y_t(0) = g(t) + \varepsilon_t$ for $t = 1, \dots, n_{\text{pre}} + 12$, with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ for $t = 1, \dots, n_{\text{pre}} + 12$. Treatment begins at $t_0 = n_{\text{pre}} + 1$, and a constant effect τ is added over the following twelve months. We evaluate point estimation using absolute bias and RMSE, but our primary interest is interval coverage as the amount of pre-treatment history changes. The two trend-based scenarios also include a small random Gaussian bump, introducing local departures from their underlying trend functions.

We consider three forms for g , all chosen to reflect features of the application but differing in how well they align with the GP’s assumed function class (see Supplementary Material C for more details). The first, *Smooth*, constructs the trajectory from a finite expansion of the combined kernel, providing a favorable benchmark in which the DGP is closely aligned with the estimator. The GP yet selects its own hyperparameters and is not given the generating specification. The second, *Nonlinear trend*, combines relatively mild seasonality with a trend that initially rises but gradually approaches a plateau. The third, *Saturating trend*, combines a trend that flattens at both ends with a stronger annual cycle, requiring the estimator to extrapolate both the longer-run movement and pronounced seasonal variation.

We fit the GP using the same specification as in Section 6 and compare it with three commonly used alternatives: (i) segmented regression with a linear pre-trend (and annual harmonic), (ii) C-ARIMA (Menchetti et al., 2023), and (iii) CausalImpact, a Bayesian structural time-series model (Brodersen et al., 2015). For each scenario, n_{pre} ranges from 12 to 120 months in twelve-month increments, with 500 replications at each length. We report absolute bias and RMSE for point estimation and 95% coverage for interval calibration over the post-treatment window.

Results. Figure 2 shows the clearest differences in interval coverage rather than in point estimation. The GP maintains coverage at or above the nominal 95% level across all three scenarios, including when only twelve pre-treatment observations are available. This is consistent with the theoretical argument in Section 4: as the counterfactual becomes harder to determine from the observed history, the GP’s uncertainty can expand to reflect the range of continuations allowed by

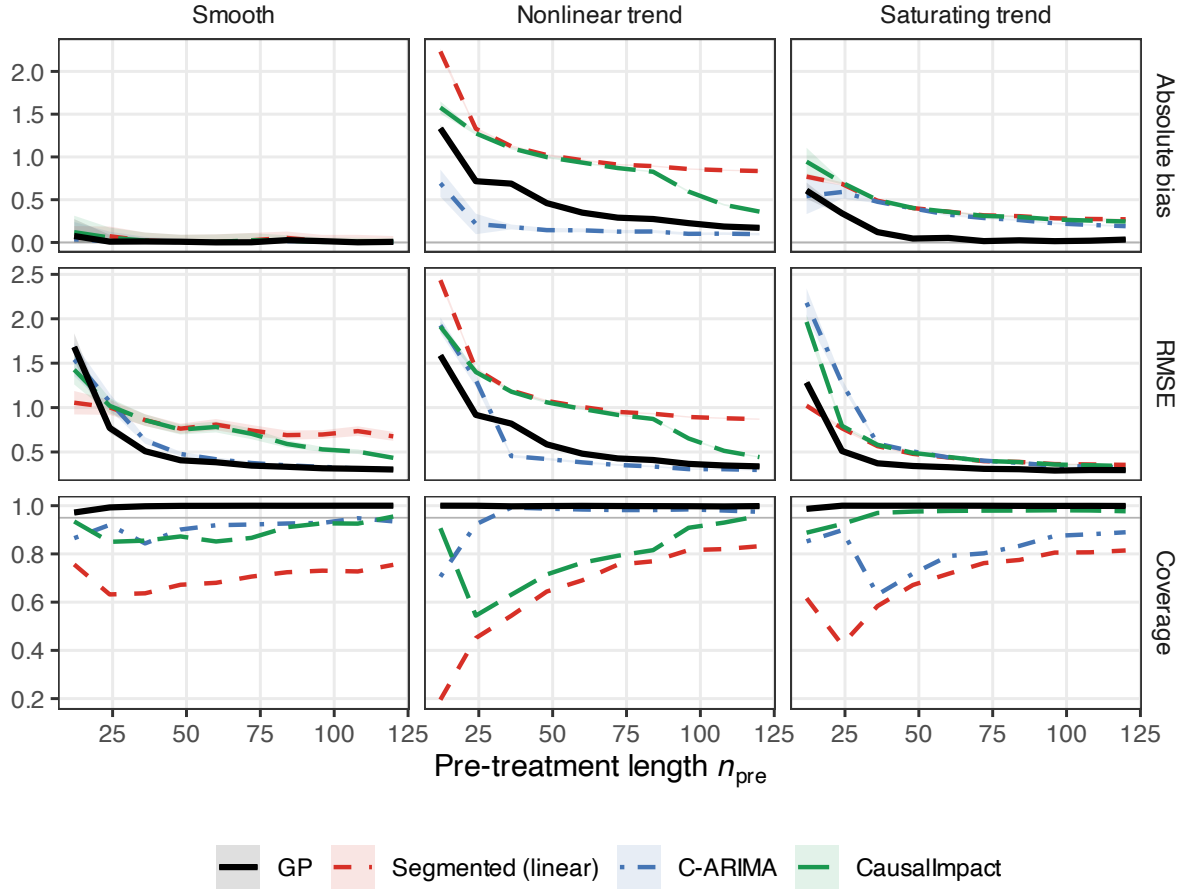


Figure 2. Absolute bias, RMSE and 95% coverage against the pre-treatment length n_{pre} , over a fixed twelve-month horizon and 500 replications, for three scenarios (columns). Bias is that of the average post-treatment effect; RMSE and coverage are computed at each post-treatment period and then averaged. The GP (black) is compared with a segmented regression, C-ARIMA and CausalImpact. Shaded bands are Monte-Carlo 95% intervals.

the kernel. The benchmark methods behave differently. Segmented regression substantially under-covers, particularly when the untreated trajectory departs from a linear trend; its intervals reflect uncertainty within the fitted specification but not disagreement among alternative continuations. C-ARIMA, with the seasonal period specified, approaches nominal coverage as the pre-treatment history grows, but under-covers with shorter histories. CausalImpact shows less stable coverage across sample sizes, falling below nominal at intermediate lengths before improving with additional pre-treatment observations. By contrast, the differences in absolute bias and RMSE are considerably smaller, and point estimation generally improves for all methods as more pre-treatment

observations become available. The main advantage of the GP in these simulations is therefore not uniformly more accurate point prediction, but more reliable uncertainty quantification across the extrapolation settings considered.

6 Empirical Application

Did the Supreme Court’s decision in *District of Columbia v. Heller* (2008) affect legal handgun purchases? On June 26, 2008, the Court recognized an individual right to keep and bear arms under the Second Amendment and struck down D.C.’s handgun restrictions. Although the ruling’s immediate regulatory effect was concentrated in D.C., its broader constitutional significance extended beyond the District. The decision established a major precedent concerning the individual right to bear arms and could have influenced expectations about future gun regulation and purchasing behavior nationwide, even though the application of the Second Amendment to the states was not established until *McDonald v. Chicago* (2010).

This makes *Heller* a useful setting for the design developed above. Other jurisdictions experienced no comparable immediate regulatory change, but their purchasing behavior may nevertheless have responded to the ruling. They therefore cannot simply be regarded as clean unaffected controls. The empirical question is thus not simply treated versus control, but where and when the ruling produced a detectable change. If the unit-specific counterfactuals are well calibrated, we should see a large immediate effect in D.C. and little corresponding movement elsewhere.

We use monthly handgun background check rates per 100,000 population from the FBI’s National Instant Criminal Background Check System (NICS) as a proxy for handgun purchases (Steidley and Kosla, 2018). The panel includes all U.S. states and D.C. except Hawaii, whose record is incomplete, yielding $N = 50$ jurisdictions. The raw monthly series appears in Supplementary Material E. We restrict the post-treatment analysis to July–October 2008, ending before the November presidential election. This choice reflects the possibility that election-related concerns about future gun restrictions could influence purchases, making it more difficult to isolate the effect of *Heller*. Anticipation of the ruling itself presents a separate concern (Bertoli et al., 2026). In D.C., the near-total handgun ban limited opportunities for legal purchases in advance of the decision, lend-

ing some support to the no-anticipation assumption. Supplementary Material D.3 examines these identifying assumptions in greater detail.

Model. The monthly background check series exhibit seasonality as well as longer-run movements, motivating the Gaussian+Periodic+Linear kernel described in Section 3.2. We fit a separate GP for each jurisdiction, allowing the Gaussian component to capture local variation, the periodic component to represent annual seasonality ($p = 12$), and the linear component to accommodate longer-run trends. Month indicators are also included as covariates to account for month-specific baseline levels.

Illustrative Cases: D.C. and Washington State. We begin with D.C., where the ruling directly invalidated the handgun ban, and Washington state, whose pre-treatment trend and seasonal patterns closely resemble the national average. Figure 3 compares their fitted counterfactuals, post-treatment effects, and placebo estimates. The contrast is clear in both the fitted counterfactuals and the resulting effect estimates. In D.C., observed handgun checks rise sharply above the predicted counterfactual immediately after the ruling, producing positive estimated effects throughout the four-month post-treatment window. The placebo estimates remain near zero. Washington state shows no comparable departure: its post-treatment observations remain within the predicted counterfactual range, and the estimated effects are statistically indistinguishable from zero.

Localized *Heller* Effect and National Average. Figure 4 extends the analysis to all jurisdictions. The left panel presents their four-month cumulative effects, standardized by each jurisdiction’s pre-treatment variability to facilitate comparison across different baseline purchase levels. D.C. stands out with a large positive standardized effect, whereas the other jurisdictions cluster around zero. The standardized magnitude for D.C. should be interpreted cautiously, however. Its handgun ban kept pre-treatment checks close to zero, leaving little variation against which to scale the estimated effect. We therefore report its magnitude in the original units: the cumulative four-month effect is 15.1 additional checks per 100,000 population, with a 95% interval of [13.0, 17.3], equivalent to roughly 90 additional background checks.

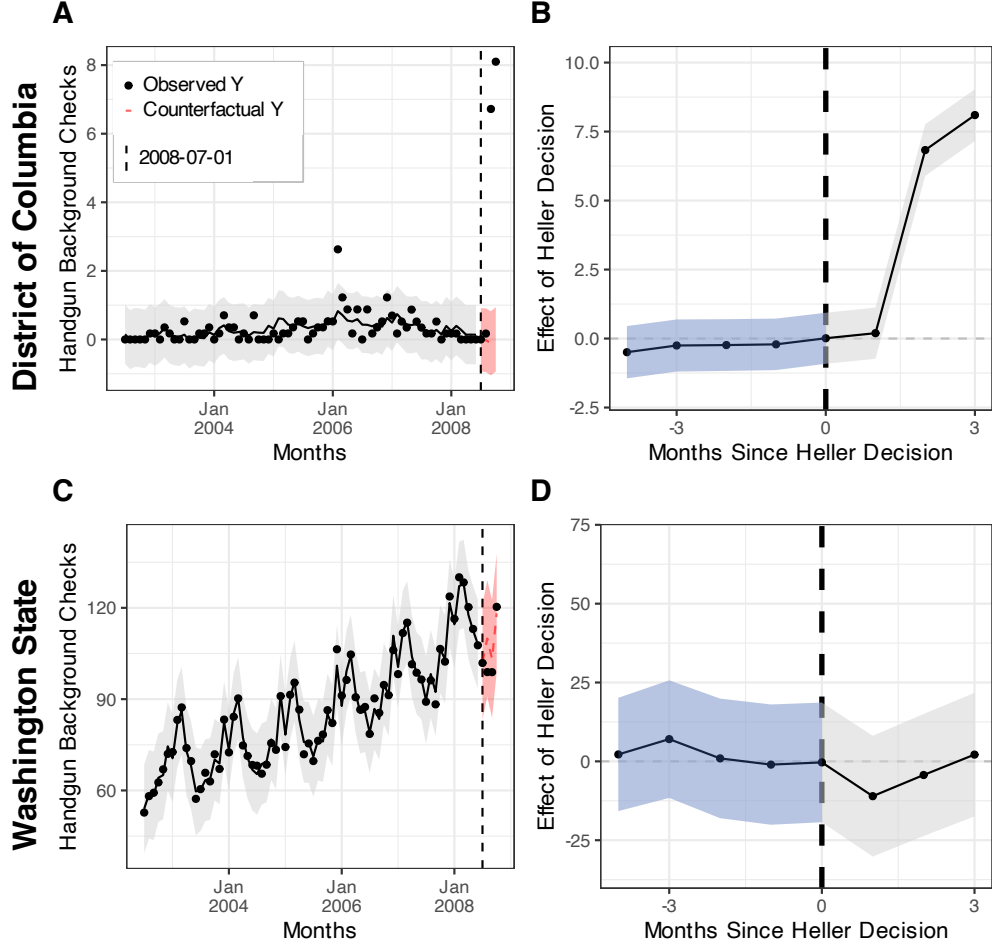


Figure 3. Counterfactual Predictions and Treatment Effects for D.C. and Washington State. *Note:* The top row shows D.C. and the bottom row Washington state. The left panels display observed outcomes (black dots), the fitted GP (gray bands), and post-treatment counterfactual predictions (red bands). The right panels show monthly treatment effects (gray) and placebo estimates (blue). Vertical dashed lines indicate the *Heller* decision.

The right panel traces the national average effect $\hat{\tau}_{t^*}$ over time. It remains close to zero across the post-treatment months, and the cumulative version in Supplementary Material E shows the same pattern. Together with the jurisdiction-specific estimates, this suggests that the observed increase was concentrated in D.C. rather than broadly shared across the country. The national results also provide context for the concern about election-related purchasing. A widespread response to the approaching election could have produced increases outside D.C., yet we find little corresponding movement in the national average during July–October. This pattern is less consistent with a broad nationwide increase, although it cannot rule out election-related behavior specific to D.C. or other

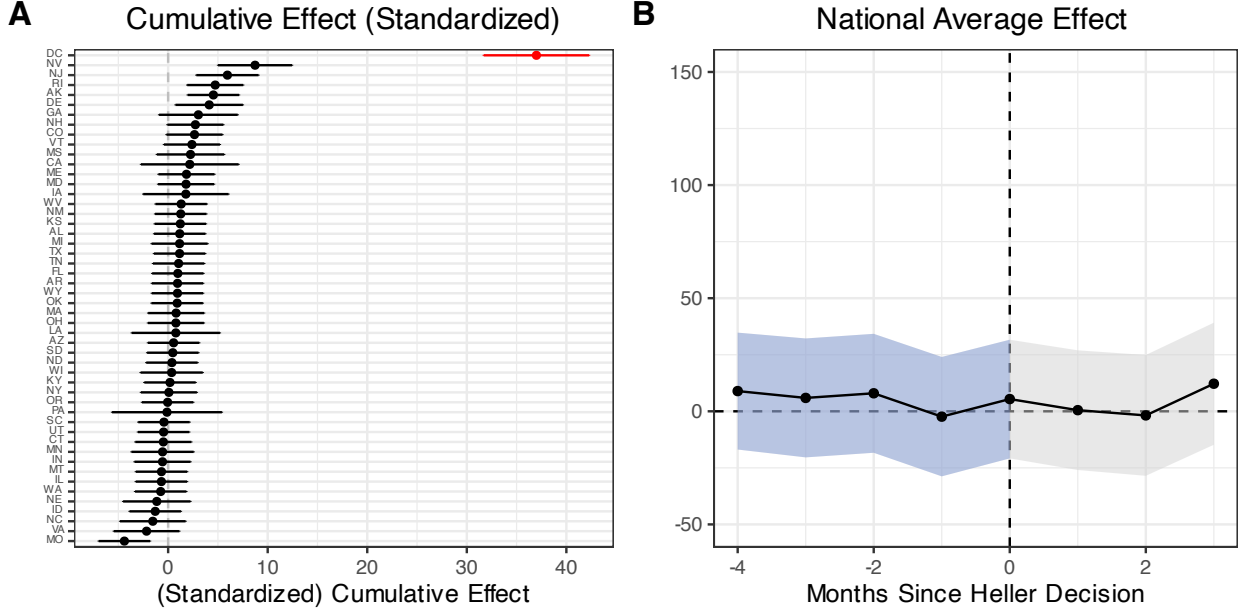


Figure 4. Heterogeneous and Nationwide Effects of the *Heller* Decision on Handgun Purchases. *Note:* The left panel gives unit-specific standardized cumulative effects for the four months following the decision, with 95% prediction intervals. The right panel gives the estimated national average treatment effect $\hat{\tau}_{t*}$ by month.

concurrent influences on handgun purchases there.

We further assess the counterfactual predictions using the one-step-ahead placebo check of Section 3.3. Segmented regression excludes zero in 58–67% of pre-treatment periods, indicating apparent effects where treatment had not occurred, whereas the GP excludes zero in none. The contrast mirrors the simulations of Section 5: segmented-regression intervals remain conditional on the chosen trend specification and do not account for alternative trajectories that fit the pre-treatment record similarly well but diverge out of sample. The GP placebo results are therefore reassuring about its pre-treatment forecasting performance, although they cannot prove that identifying assumptions hold. Additional diagnostics support the stability of the reported estimates. In-sample residuals show no visible deterioration as the treatment date approaches, and sensitivity analysis across a broad range of length-scale values b yields similar estimates for the D.C. effect and the national average (Supplementary Materials E and F).

We also examine long gun checks as a placebo outcome, since the ruling directly invalidated D.C.’s handgun ban rather than a comparable restriction on long guns. The standardized cumulative effect in D.C. is 0.25, with a 95% interval of $[-5.2, 5.7]$, providing no evidence of a comparable

change in this outcome. Supplementary Material E presents the cumulative effects, placebo comparisons, residual diagnostics, and long gun results in greater detail.

7 Relation to Existing Approaches

Methods for constructing counterfactual outcomes differ in where they obtain the information needed to predict what would have happened without treatment. Some rely primarily on contemporaneous comparisons across units, others on a unit’s own history over time, and still others combine the two. When treatment occurs simultaneously across all units, contemporaneous untreated comparisons are no longer available, making the unit’s pre-treatment history the primary source of information about its counterfactual. The central question then becomes how that history is projected forward and, critically, how uncertainty about that projection is calibrated. Viewed this way, our approach is closest to methods that learn the untreated process from pre-treatment data, but differs in the assumptions placed on its continuation and in how uncertainty about extrapolation is characterized.

Time Series Approaches. Classical ITS methods, from the intervention analysis of [Box and Tiao \(1975\)](#) to segmented regression, typically specify both a model for the untreated trajectory and a form for the treatment response, such as a step, pulse, or gradual change. These choices make estimation and interpretation straightforward, but they also restrict both how the untreated outcome may evolve and, in many specifications, how the treatment effect may unfold over time. More recent approaches relax parts of this structure. Within a potential outcomes framework, [Miratrix \(2022\)](#) simulate counterfactual trajectories from a linear prediction model, while [Felton \(2023\)](#) imputes counterfactual outcomes from ARIMA fits. Our approach shares their basic strategy of learning the untreated process from the pre-treatment history. The distinction lies in how the continuation is modeled: the GP uses a covariance kernel to describe the class of trajectories regarded as plausible, and its posterior uncertainty changes with how strongly the observed history constrains the counterfactual at the prediction point.

The comparison with `CausalImpact` ([Brodersen et al., 2015](#)) is particularly useful in this re-

spect. It uses a Bayesian structural time-series model with latent state components, and, as [Hazlett and Xu \(2018\)](#) emphasize, its counterfactual can be expressed as a linear function of prior outcomes. The GP shares this feature: its posterior mean is also a linear smoother, $\hat{g}(\mathbf{X}_{t^*}) = \sum_t w_t Y_{it}$. Thus, the relevant distinction is not whether the prediction is linear in the observed outcomes, but how the weights and the underlying trajectory are constructed. In the GP, the kernel determines both the prediction weights and the covariance structure linking the observed pre-treatment inputs to the post-treatment target. Its posterior variance, in turn, reflects how much of that target remains unresolved by the observed history, which is the connection to extrapolation developed in [Section 4](#).

C-ARIMA ([Menchetti et al., 2023](#)) is similarly close in spirit. It estimates the untreated process from the pre-treatment series and forecasts that process forward, rather than imposing a prespecified treatment response through a regression fitted to the full series. The difference is again in the model used to extend the observed history: C-ARIMA represents temporal dependence through autoregressive, differencing, and moving-average components, whereas the GP represents it through a covariance kernel. Both approaches therefore quantify forecast uncertainty under assumptions about how the untreated process evolves. Our contribution is to make the GP version of those assumptions explicit through its associated function space and to connect the resulting posterior variance to worst-case learning error within that class. The choice between these approaches ultimately depends on which description of the untreated trajectory is substantively plausible for the application at hand.

Related Identification Strategies. The identifying logic also connects to methods that construct counterfactuals from relationships learned before treatment. DiD and its modern variants use changes in untreated comparison units to infer how treated outcomes would have evolved absent treatment, under a parallel trends assumption. Synthetic control instead constructs a weighted combination of untreated donor units that reproduces the treated unit’s pre-treatment trajectory and relies on that relationship to remain informative after treatment. When all units are treated simultaneously, neither source of contemporaneous untreated information is available. Our approach therefore shifts the relevant comparison from across units to within a unit over time.

This shift changes, rather than removes, the identifying requirement. Mean Sufficiency limits

the extent to which unobserved factors can alter the conditional mean once the observed covariates are given, while the kernel governs how the relationship learned over the observed history is allowed to continue beyond it. In this sense, temporal extrapolation takes the place of the cross-unit comparisons available in DiD and synthetic control, and the credibility of the resulting counterfactual depends on whether that extrapolation is substantively defensible.

RDiT provides a different way to use temporal information, exploiting the local comparability of observations immediately before and after a treatment cutoff to estimate an effect near the event time. This can be attractive when the causal question is itself local, but the same local comparison does not generally reveal how an effect develops, persists, or fades over a longer post-treatment period. Our framework instead constructs a counterfactual for each post-treatment period, which requires stronger extrapolation but also makes those assumptions explicit through the kernel and allows uncertainty to increase as the prediction moves away from the observed history. RDiT also faces particular challenges with discretely observed outcomes (Bertoli et al., 2026; De la Cuesta and Imai, 2016), which the GP framework accommodates directly.

8 Concluding Remarks

When treatment reaches all units at once, the usual contemporaneous sources of counterfactual information disappear, leaving the researcher to learn from what happened before treatment and extrapolate beyond it. The difficulty is not simply obtaining a flexible forecast. Different counterfactual trajectories may fit the observed history similarly well while implying quite different outcomes afterward, and inference should reflect that disagreement rather than condition entirely on a single continuation. We propose GP regression as one way to address this problem. The kernel makes explicit which kinds of untreated trajectories the model is willing to consider, while the posterior distribution describes how strongly the observed pre-treatment history constrains those trajectories after treatment.

The identification analysis separates this extrapolation problem from the assumptions needed to give the projected counterfactual a causal interpretation. The theoretical results then distinguish approximation error from the learning error that remains because a finite pre-treatment history

does not fully determine the counterfactual beyond the observed support. Within the function class induced by the kernel, the GP posterior variance has a direct interpretation in terms of this learning problem: it reflects the extent to which admissible trajectories can remain similar before treatment yet diverge afterward, together with uncertainty propagated from noisy observations. The calibrated simulations show that this uncertainty produces reliable coverage across the settings considered, and the *Heller* application illustrates how the approach can recover dynamic effects without requiring a prespecified treatment-response pattern. An open-source R implementation accompanies the paper.

The method nevertheless inherits the central limitation of any extrapolation-based design: its conclusions depend on what continuations are regarded as plausible. Here that judgment enters through the kernel and its implied function class. The worst-case learning bound also scales with the RKHS norm of the target, $\|g^*\|_{\mathcal{H}_k}$, which cannot generally be learned from the observed data alone. More practically, fitting each unit separately does not exploit possible cross-unit structure, potentially sacrificing precision when individual pre-treatment series are short. Hierarchical or multi-task formulations (Ben-Michael et al., 2023) could borrow information across units, although doing so would introduce additional assumptions about their similarity. Computation can also become burdensome for very long series because standard GP fitting requires inversion of the kernel matrix; sparse approximations (e.g., Titsias, 2009) provide one possible extension.

Several extensions merit exploration. Because covariance is defined by distances between inputs rather than by a requirement of equally spaced observations, the framework can accommodate irregular observation times, which arise frequently in administrative and judicial data; a fuller study of performance under irregular spacing and missing observations would be useful. More broadly, the argument developed here is not specific to ITS design. Whenever a counterfactual must be learned from observed history and then carried beyond it, some assumption must determine which continuations remain plausible. The benefit of the GP is not that it removes this requirement, but that it makes the assumption explicit and provides a principled way to *price* the extrapolation uncertainty that remains.

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